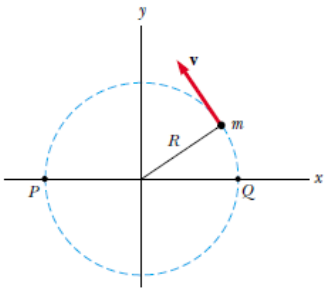
PHY101: Introduction to Physics-1

Monsoon, 2018

Tutorial-10

1. A particle of mass m moves in a circle of radius R at a constant speed v, as shown in figure. If the motion begins at point Q at time $t = 0$, determine the angular momentum of the particle about point P as a function of time.



2. Three particles of masses 3, 5 and 2 units are located at points (-1, 0, 1), (2, -1, 3) and (-2, 2, 1) respectively. Find the moment of inertia of the system of the particles about the y-axis.

3. A uniform solid cylinder has a radius R, mass M, and length L. Calculate its moment of inertia about its central axis. If the length of the cylinder is increased to 2L, while the mass M and radius R are held fixed, how does that change the moment of inertia of the cylinder?

4. Find the moment of inertia of a thin sheet of mass M in the shape of an equilateral triangle around an axis through a vertex, perpendicular to the sheet. The length of each side is L.